

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF ECONOMICS Data, research design
and descriptive statistics

Week 1 · Lectures 1 and 2

Tashkent International University · Faculty of Economics

2025/2026 academic
year



Why is this course? Every empirical claim relies on a statistical conclusion

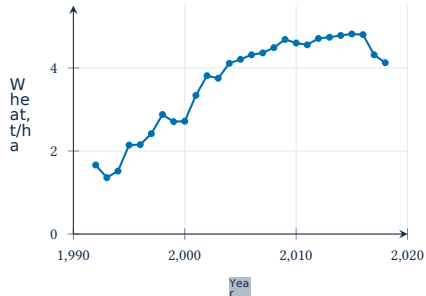
In 2019, a newspaper wrote:

“Wheat productivity in Uzbekistan has increased dramatically since independence — agricultural reforms are a success.”

This sentence contains four statistical claims:

- ▶ “yield” — how was it measured?
- ▶ “increased dramatically” — by how much? Over what period?
- ▶ “successful” — where is the basis for comparison?
- ▶ “because of reforms” is a causal claim

Audience Question Which of the four claims does this graph support?



Source: FAOSTAT / Our World in Data

Learning Objectives for Week 1

By the end of this week, you will:

1. Explain the structure of a data table: unit of observation and variable.
2. Classify variables by type.
3. Distinguish between observational research and experiment; identify confounding factors.
4. Calculate measures of center and dispersion and justify when each is appropriate.
5. Choose the correct graph based on the type of variable.
6. Identify a confounding graph and explain its error.

Learning Outcome CLO1. Summarize and visualize data with appropriate descriptive indicators and graphs (Bloom: Apply).

Today's data set: central-asia-yield.csv — Yields of five crops in Uzbekistan, Kazakhstan, Kyrgyzstan, Tajikistan, and Turkmenistan, 1992–2018 (135 rows, real data).

Semester Map: Where We Are



we are here

Why this week is important. Descriptive statistics is the alphabet for the next 15 weeks. You can't understand the central limit theorem or regression without understanding the mean and standard deviation.

A common mistake But be careful: descriptive statistics describe, not infer. "The mean in this sample is 4.6" $\neq$ "the mean in the population is 4.6." This distinction becomes a central theme in Week 7.

What is information?

Anatomy of a data table

country	year	wheat	potato	barley
Uzbekistan	1992	1.662	8.51	1.410
Uzbekistan	2018	4.126	24.42	1.703
Kazakhstan	2018	1.116	19.79	1.232
Tajikistan	2018	3.147	27.90	2.013

→Row =

one observation unit ↓ Column
= one variable

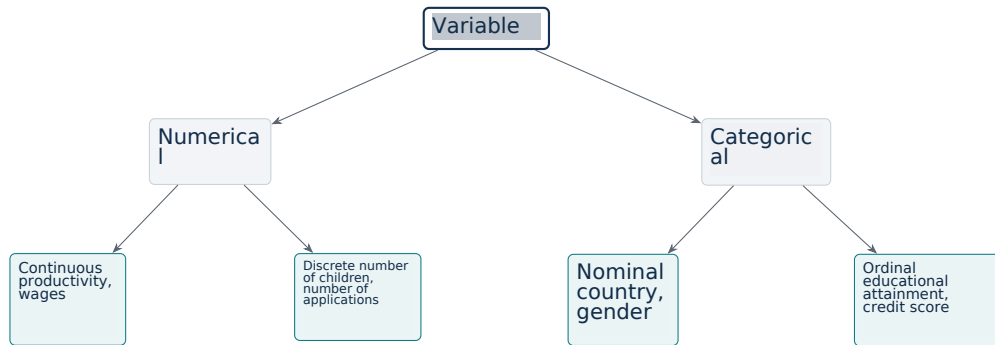
Definition Observational Unit — an object about which information is collected. Here: the state of a single country in a single year.

Variable — a characteristic measured in terms of observation units.

Observation — all the values of the variables for one unit (one row).

Audience Question If the unit of observation is “country”, how many rows are left in the table and what does the wheat column mean?

Taxonomy of variables



Common mistake Trap. Just because it is written as a number does not make it numerical. A zip code of 100,000 is not a number, it is a nominal label. Calculating its mean is meaningless.

Definition Why is it important? The type of variable determines all subsequent decisions: which indicator, which graph, which test.

Audience Exercise: Classify Think

— Match — Tell (2 minutes)

Variable Type? Appropriate indicator?

Student's test score (0-100)

Student's region of residence

Number of cars in the family

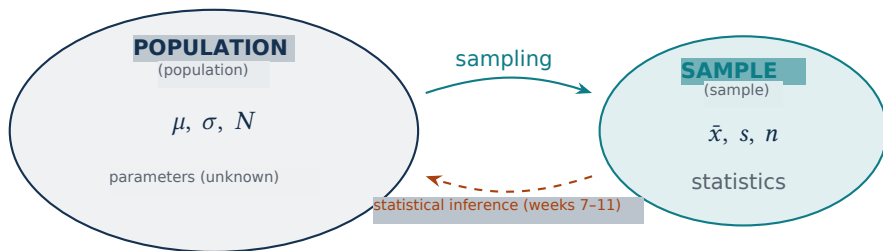
Credit rating (A, B, C, D)

Waiting time in line at a bank

Phone number

Audience Question The last two are the most controversial. Why?

Population and sample



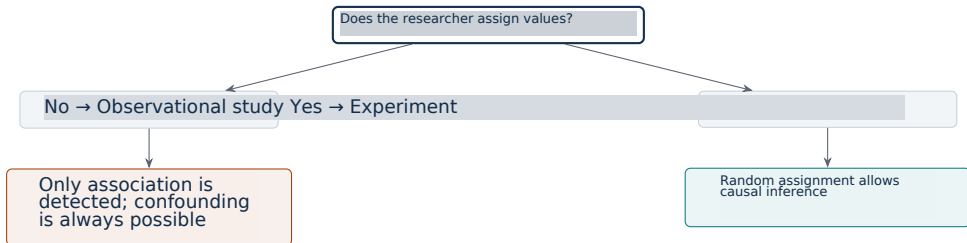
Definition Parameter — a numerical characteristic of the population (μ, σ, N). Usually unknown.

Definition Statistic — a number calculated from the sample ($\bar{x}, s, n$). It turns out differently for each sample.

Common mistake Greek letters → population; Latin letters → sample. This rule does not change throughout the course.

Research design

Where does the data come from?



Our data. The productivity data are observational. No one randomly assigned countries to “reform” or “no reform” status.

A common mistake Therefore, the conclusion that “reforms increased productivity” does not follow from this data. We will return to this issue in detail in Week 13.

Sampling Strategies

Simple random Stratified Clustered Systematic



Method Idea When preferred

Simple random Each unit is selected with equal probability When the list is complete
Stratified The set is divided into homogeneous groups and one is selected from each When the difference between the groups is large

Clustered Entire groups are selected at random When geographically dispersed
Systematic Every k th unit is selected When there is a ranked order

Sampling Bias: The Most Expensive Statistical Error

Definition Sampling Bias — a systematic misrepresentation of the sample population.

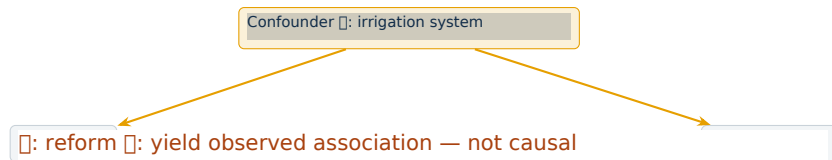
Three classic types:

- ▶ **Convenience Bias** — whoever is close is asked.
- ▶ **Non-Response Bias** — those who respond to a survey are systematically different from those who do not.
- ▶ **Survivor Bias** — only “survivors” are shown.

A common mistake Large sampling bias does not cure. A 1936 Literary Digest survey collected 2.4 million responses and incorrectly predicted the outcome of the US election. The reason: the list was compiled from telephone and car owners.

Increasing n reduces random error, not systematic bias. This is a distinction that comes up again and again in the course.

Confounder Why observational data does not prove causality



Definition Confounder — a third variable that affects both $\square$ and $\square$. It creates a non-causal relationship between $\square$ and $\square$.

Audience Question What other factors affect both reform and yield? Name at least three.

Random assignment “disconnects” $\square$ from $\square$ — on average $\square$ is the same in both groups.

Four obligations in data collection:

1. Informed consent — the respondent must know why the data is being collected.
2. Confidentiality — personally identifiable information is protected.
3. Anonymization — removing the name is not enough; several variables together can reveal the identity.
4. Non-use beyond the intended purpose — information collected for one purpose is not transferred to another.

Common mistake Re-identification. Just a zip code + date of birth + gender can be enough to uniquely identify most people — even if the name is not.

In a semester project, you will openly report the source of the data, the method of obtaining it, and the issue of confidentiality. This is evaluated.

Center measurements

Definition Mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Median — the middle value of a sorted set.

Mode — the most frequently occurring value.

Which one when?

Mean Symmetrical distribution; needed for further calculations
Median Skewed distribution; there are outliers
Mode Categorical variable; the “most typical” value

Common mistake Only the mode is meaningful for a categorical variable. There is no such thing as a “mean range.”

Important property. $\sum (x_i - \bar{x}) = 0$ — the sum of the deviations from the mean is always zero. This is the reason for the definition of variance: deviations cannot simply be added.

Let's calculate manually Uzbekistan, wheat yield (t/ha), 2010-2018

Year	
2010	4,600
2011	4,556
2012	4,710
2013	4,739
2014	4,782
2015	4,818
2016	4,802
2017	4,316
2018	4,126
$n = 9 \quad \sum x_i = 41,449$	

1. Mean

$$\bar{x} = \frac{41,449}{9} = 4.605 \text{ t/ha}$$

2. Median — sort:

4,126 4,316 4,556 4,600 4,710 4,739 4,782 4,802 4,818

$n = 9$ odd $\Rightarrow$ 5th value: Med = 4.710

3. Comparison

$$\bar{x} = 4.605 < \text{Med} = 4.710$$

Common error The mean is less than the median — the distribution is skewed to the left. Reason: Low values in 2017-2018 pull the mean down.

Why is the median stable? Let's try changing one value

Let's change the 2015 value from 4.818 to 12.0 (for example, the unit of measurement was entered incorrectly):

With Original Error			
Mean	4.605	5.403	Median
	4.710	4.710	
Change — $\Delta \bar{x} = +0.798$ $\Delta \text{Med} = 0.000$			

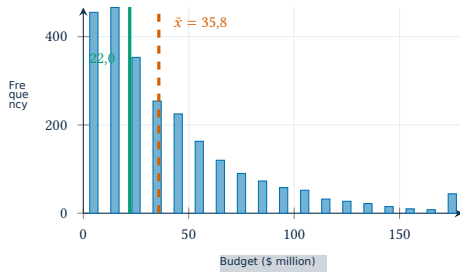
Definition Robustness — the indicator's low sensitivity to outliers.

The median is stable, the mean is not stable.

Audience Question If this 12.0 is not an error, but a real value — which indicator would you report? Why?

Practical rule. Calculate both. The difference itself is information about the shape of the distribution.

Skewed distribution: real example 2,467 films,
production budgets, 2000–2018



n = 2,467
 Mean \$35.8 million
 Median \$22.0 million
 IQR \$10.0 million
 Range \$1.0 - \$350.0

This is a typical shape of the distributions of income, wealth, city population, and firm size.

Audience Question: “average salary”

Audience Question In a company, 9 employees earn 5 million soums per month, and the director earns 100 million soums.

(a) What is the average salary? (b) What is the median salary? (c) What number is used in company advertising? What number should a job seeker know?

Audience Question: “average salary”

Audience Question In a company, 9 employees earn 5 million soums per month, and the director earns 100 million soums.

(a) What is the average salary? (b) What is the median salary? (c) What number is used in company advertising? What number should a job seeker know?

Solution

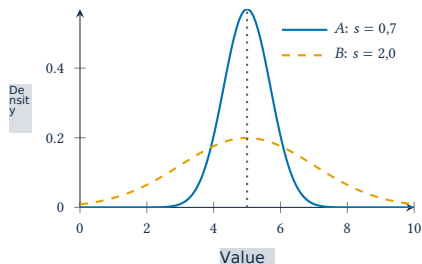
$$\bar{x} = \frac{9 \times 5 + 100}{10} = \frac{145}{10} = 14.5 \text{ million}$$

Median: In a ranked list, the 5th and 6th values are both 5 $\Rightarrow$ Med = 5 million.

A common misconception is that 9 out of 10 employees earn less than the “average.” Statistics are not false — but the statistics chosen are misleading.

Dispersion measures

Average is not enough



Both have $\mu = 5$

Three measures of dispersion

Definition Range = $\max - \min$ — simple but relies on only two values.

Interquartile Range IQR = $Q_3 - Q_1$ — stable.

Standard Deviation

$$s = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$$

Audience Question Two provinces have the same average income, but s differs. Which province has higher inequality? Which number is more important for policy?

Variance: why square, why $n-1$?

Step 1 — deviations

$$d_i = x_i - \bar{x}, \quad \sum_{i=1}^n d_i = 0$$

Therefore, they cannot simply be added.

Step 2 — squaring

$$\sum d_i^2 \geq 0$$

Squaring: (i) loses sign, (ii) amplifies large deviations, (iii) differentiable — necessary for the next theory.

Step 3 — averaging

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Theorem Why $n-1$? The deviations are taken from the sample mean, not the population mean. Since $\bar{x}$ itself is calculated from the data, the deviations are “too small”.

The condition $\sum d_i = 0$ loses one degree of freedom: if $n-1$ deviations are known, the last one is determined automatically.

Dividing by $n-1$ makes s^2 an unshifted estimator for σ^2 —we will prove this in Week 9.

Let's calculate by hand: s^2 and s
 Those 9 values, $\bar{x} = 4.605$

Year	x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
2010	4,600	-0,005	0,0000
2011	4,556	-0,049	0,0024
2012	4,710	+0,105	0,0110
2013	4,739	+0,134	0,0179
2014	4,782	+0,177	0,0313
2015	4,818	+0,213	0,0452
2016	4,802	+0,197	0,0387
2017	4,316	-0,289	0,0836
2018	4,126	-0,479	0,2296
			$\approx 0,000$
			0,4599

$$s^2 = \frac{0,4599}{9 - 1} = \frac{0,4599}{8} = 0,0575$$

$$s = \sqrt{0.0575} = 0.240 \text{ t/ha}$$

Coefficient of variation

$$CV = \frac{s}{\bar{x}} = \frac{0,240}{4,605} = 0,052 = 5,2\%$$

CV is unitless — so it is useful for comparing variables in different units.

A common error is that $\sum (x_i - \bar{x})$ is not exactly zero — this is a rounding error. Round off at the end of the calculation.

Comparison: which country is "stable"? Wheat yield, 1992-2018, $n = 27$

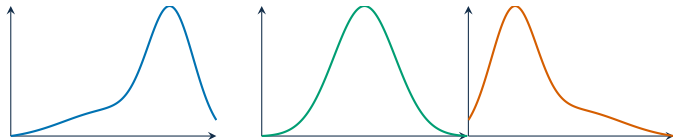
Country	Med	IQR	CV, %		
Uzbekistan	3.605	4.126	1.179	1.938	32.7
Tajikistan	1.991	1.987	0.853	1.805	42.8
Turkmenistan	1.906	1.600	0.808	1.277	42.4
Kyrgyzstan	2.235	2.317	0.277	0.315	12.4
Kazakhstan	1.027	1.076	0.260	0.359	25.3

Common mistake Important warning. Kazakhstan has the lowest yield — but it is the largest wheat exporter in the region. Yield (t/ha) $\neq$ total yield (tons). Always check what the indicator measures.

Distribution form

Skewness and Modality

Left-skewed Symmetric Right-skewed



Shape Relationship

Left-skewed $\bar{x} < \text{Med}$

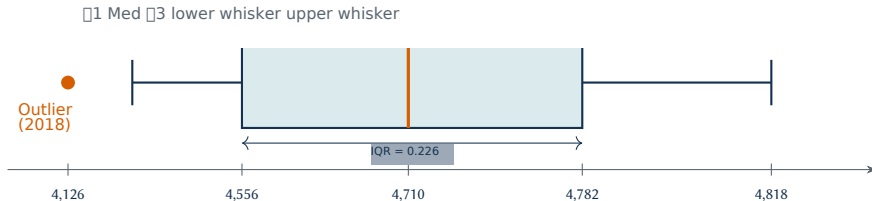
Symmetric $\bar{x} \approx \text{Med}$

Right-skewed $\bar{x} > \text{Med}$

Common Mistake This is not a rule, but a tendency. There are artificial examples where $\bar{x} < \text{Med}$ is strongly skewed. The final conclusion about the shape is drawn from the graph.

Modality. A bimodal distribution often indicates that two different groups are mixed in the data — check this out.

Five-point summary and boxplot



Outlier rule ($1.5 \times \text{IQR}$)

$$\square 1 -1.5 \text{ IQR} = 4.556 - 0.339 = 4.217$$

$$\square 3 + 1.5 \text{ IQR} = 4.782 + 0.339 = 5.121$$

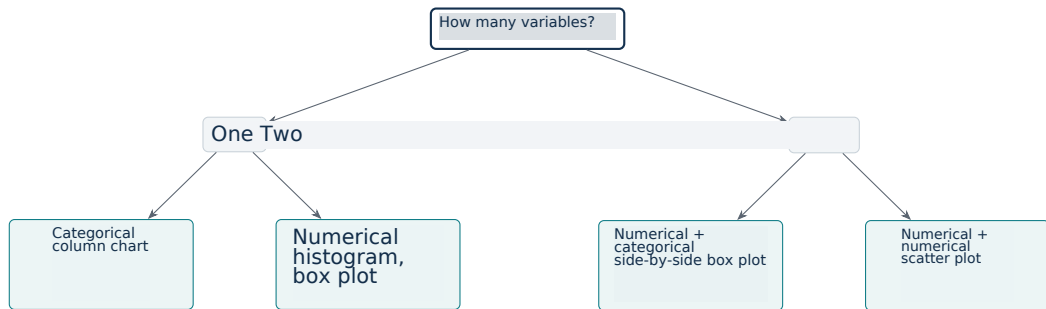
$4.126 < 4.217 \Rightarrow 2018$ is defined.

Common mistake “Outlier”
≠ “error”. 2018 was a drought year
— that’s the real value. Deleting it
would be data corruption.

A rule is a statement of intent, not a
judgment. Always ask why.

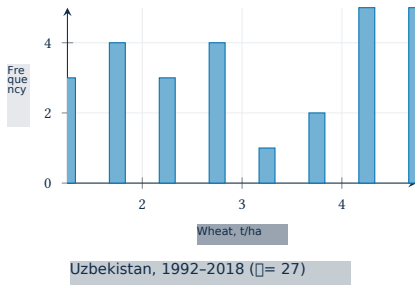
Visualization

Which graph should I choose?



Basic rule. The type of graph is determined by the variable type — not its beauty. For two categorical variables: a bar chart with proportions or a frequency table (this leads to the chi-square test in week 11).

Histogram and boxplot: what do they show?



A histogram shows:

- ▶ shape (here: two peaks)
- ▶ gaps and clusters
- ▶ whole distribution

A boxplot shows:

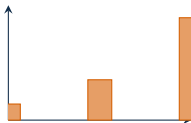
- ▶ quintile conclusion
- ▶ outliers
- ▶ comparison of groups

A common mistake Boxplot hides two peaks. When shape is important, a histogram is.

Question: why is the distribution two peaks? Hint: this is a time series.

Statistical Ethics II: Misleading with Graphs

1. Crossed axis



Does not start at 0 — the difference is 6 times greater

It seems

2. Straight axis



Same data — the real difference is small

Three classic tricks:

1. Crossed $\square$ -axis — makes a small difference look large.

2. Incorrect aggregation — combining groups and drawing the opposite conclusion (Simpson's paradox, week 13).

3. Two $\square$ -axes — you can choose the scale and make any “relationship” you want.

A common mistake Ethical rule. In a bar chart, the $\square$ -axis always starts at zero. This is not necessary in a line graph — but the scale should be clearly indicated.

**Summar
y**

Common Student Mistakes and Misconceptions

Misconception Why it's wrong

“The mean is the best measure of center.” In a skewed distribution, the mean does not represent the typical value. In a sample of 10, 9 people score below the “mean.”

“You should eliminate the outlier.” An outlier is a consequence of the $1.5 \times \text{IQR}$ rule, not a judgment. The 2018 drought is a real event.

“A larger sample is always better.” Increasing n reduces random error, not bias. Literary Digest: 2.4 million answers and an incorrect prediction.

“Written in numbers, it means it’s a numerical variable.” Zip code and phone number are nominal labels. Their average is meaningless.

“ σ large — so the data is bad.” σ — a measure of natural variation, not a quality assessment. A high σ may be a meaningful finding (e.g., inequality).

“ ρ appears to be a correlation — so the cause is found.” Confounding is always possible in observational data. Week 13 is about that.

Basic conclusions

1. The type of variable determines all subsequent decisions: indicator, graph, test.

2. How the data is collected determines what conclusions can be drawn from it. Observation $\rightarrow$ association; experiment $\rightarrow$ causality.

3. Give the center, dispersion, and shape together. Mean alone is an incomplete report.

4. For skewed data, prefer the median and IQR; for symmetric data, $\bar{x}$ and s .

5. Choose a graph based on the type of variable; do not intersect the y axis.

6. For each indicator, ask: What is it actually measuring? Yield $\neq$ yield.

Next week, Week 2 — Probability Fundamentals. We move from descriptive statistics to the language of uncertainty: elementary event space, axioms, rules for addition and multiplication, independence.

This week's assignments

Assignment Content Due Date

Required Reading IMS 2e, Chapters 1-2 (Hello data; Study design); Newbold, Chapters 1-2 until Lecture 2

Laboratory W01-lab — R or Python (both given) In the laboratory Homework W01-homework — 6 problems, 40 points Week 2, Monday Problem Set 1 MT1 is given Submit: Week 2 Project Phase 1 Verified Research Question (150 words) Week 2, Friday SI Assignment Analyze the SI descriptively and find the error in it Week 2, Friday

Common Error Academic Honesty. Homework — individual. If you use artificial intelligence, disclose at the end: which tool, what query, how you verified the result. Disclosure is not punishable; concealment is.

1. Cetinkaya-Rundel, M. & Hardin, J. (2024). Introduction to Modern Statistics, 2nd ed. OpenIntro. Chapters 1–2. openintro-ims2.netlify.app
2. Newbold, P., Carlson, W. & Thorne, B. Statistics for Business and Economics. Pearson. Chapters 1–2.
3. McClave, J. T. & Sincich, T. (2018). Statistics, 13th ed. Pearson. Chapters 1–2.
4. Heumann, C., Schomaker, M. & Shalabh (2016). Introduction to Statistics and Data Analysis. Springer. Chapters 2–3.
5. Anscombe, F. J. (1973). Graphs in Statistical Analysis. *The American Statistician*, 27(1), 17–21.
6. Rosenbaum, P. R. (2017). Observation and Experiment. Harvard University Press. Chapter 1.
7. Reference: Ritchie, H., Rosado, P. & Roser, M. Agricultural Production: Crop Yields. Our World in Data (main source: FAOSTAT). ourworldindata.org
8. Interactive: Seeing Theory, Brown University — seeing-theory.brown.edu